The Consolidation Flywheel: A Unified Theory of Addressability, Routes, and Load Factor

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November 3, 2026

Abstract

This paper synthesises six prior papers into a single dynamical theory. The claim is that logistics is not a linear value chain but a **closed loop**: addressability determines the achievable service rate; the service rate and margin determine which jobs are worth loading; the selected jobs determine route composition; route composition determines consolidation; consolidation determines contribution; and contribution determines whether the firm earns the delivery volume that generates the address data that raises addressability. The loop has two stable equilibria — a high-density equilibrium and a death spiral — separated by a separatrix. The separatrix is the same addressability threshold a^* that appeared as a static viability condition in the first paper, now reinterpreted as a dynamical boundary. The paper presents the static chain, the dynamic loop, the formal stability analysis, the seven components (each a prior paper, each an edge of the loop), the bucket threshold as an operational estimator of the drop rate, the provider assignment rule, and the operating dashboard that tells the firm which side of the separatrix it is on. The theory's practical result is a single number to watch: the ratio of observed to theoretical service rate, which diagnoses whether the corridor is addressability-constrained or consolidation-constrained, and therefore which lever to pull.

1 The Central Claim

Every prior paper in this series established one piece of a structure. This paper argues that the pieces form a loop.

The linear version of the story would be: addressability sets the service rate, the service rate sets the cost, the cost sets the price, the price sets the margin. That story is correct as far as it goes, and it is the story most logistics analysts tell. But it misses the most important thing: **the margin feeds back into the addressability**.

The mechanism is direct. Every delivery is a write to the ledger. The ledger raises addressability at the endpoint. Higher addressability raises the service rate and lowers the failure rate. Both effects raise the margin. The raised margin supports more deliveries, which generate more ledger writes, which raise addressability further.

This is not a metaphor. It is a loop with a specific gain, two specific equilibria, and a specific separatrix between them. The firm that understands the loop knows which side it is on, what moves it across, and what keeps it there. The firm that does not is running an open-loop system and will be surprised by the outcome.

The claim of this paper is that the loop is the fundamental object, and that the six prior papers are its edges.

2 The Static Chain

Within a planning period, the firm's addressability is fixed. The chain that runs within the period is:

Definition 1 (The static chain)**.** *Given a level of addressability a at each place:*

$$a \longrightarrow \mathbb{E}[\tau(a)] \longrightarrow R(a) = \frac{1}{\mathbb{E}[\tau(a)]} \quad \textit{(capacity path)} \tag{1}$$

$$a \longrightarrow \hat{p}(a),\ A(a) \longrightarrow m(a) \quad \textit{(margin path)} \tag{2}$$

$$\rho(a) = m(a) \cdot R(a) \quad \textit{(selection ratio)} \tag{3}$$

$$\rho \longrightarrow \textit{route composition} \longrightarrow N \textit{ per departure} \tag{4}$$

$$N \longrightarrow G(\textit{route}) = N \cdot \bar{m} - c_0 - \Phi C_p - c_{km} D \tag{5}$$

Three observations about this chain matter.

First, addressability enters twice, through two distinct paths. The capacity path runs through the service rate: better-addressed endpoints take less time to serve, so more drops fit in a shift. The margin path runs through success probability and surcharge: better-addressed endpoints fail less often and cost less to serve. Both paths raise the selection ratio ρ, which is why the two effects compound rather than merely add.

Second, the drop rate is a ceiling, not a result. $R(a)$ is the maximum rate at which a driver *can* serve endpoints at addressability a. The actual number of drops in a departure, N, is bounded above by $R(a) \cdot H$ but is not equal to it. The gap between the ceiling and the actual is consolidation, and consolidation is the strategic variable.

Third, the chain is not a value chain. It does not run once. It runs every period, and its output feeds the input of the next period. This is where the loop closes.

3 The Dynamic Loop

The link from contribution back to addressability is the feedback edge.

Definition 2 (The dynamic loop)**.** *Across planning periods:*

$$G > 0 \longrightarrow \textit{survival, reinvestment, growth} \tag{6}$$

$$\longrightarrow \textit{more deliveries} \tag{7}$$

$$\longrightarrow \textit{more events in the ledger} \tag{8}$$

$$\longrightarrow \textit{higher } a^D \textit{ (dynamic addressability)} \tag{9}$$

$$\longrightarrow \textit{higher } a \tag{10}$$

$$\longrightarrow \textit{higher } R(a) \textit{ and } m(a) \tag{11}$$

$$\longrightarrow \textit{higher } \rho(a) \tag{12}$$

$$\longrightarrow \textit{better route selection and consolidation} \tag{13}$$

$$\longrightarrow \textit{higher } G \tag{14}$$

The loop is closed. Contribution is not the end of the chain. It is the input to the next iteration of the chain.

Proposition 1 (The feedback is real, and it is not automatic)**.** *The loop $G \to a$ operates only if the firm **instruments every delivery**. A firm that delivers but does not record the delivery is running an open-loop system: its contribution does not raise its addressability, because the data that would raise addressability is not captured. This is the operational form of the bootstrap/moat distinction: contribution is bootstrap unless it is converted into address data.*

This is the sharpest practical result of the entire series. **Every delivery is a deposit into the ledger, or it is nothing.** The deposit is the moat. The delivery without the deposit is just a courier job.

4 The Two Equilibria

The loop has a specific dynamic structure. Its equilibria are the levels of addressability at which the loop neither amplifies nor damps.

Definition 3 (Loop gain)**.** *Let $\phi = \partial G/\partial a$ be the sensitivity of contribution to addressability, and $\psi = \partial a/\partial G$ be the sensitivity of addressability to contribution through the ledger. The* ***loop gain*** *is*

$$\Lambda = \phi \cdot \psi. \tag{15}$$

Theorem 1 (Bistability)**.** *If $\Lambda > 1$, the loop has two stable equilibria separated by a separatrix. Small positive perturbations from the separatrix grow into the high-density equilibrium; small negative perturbations shrink into the death spiral.*

Sketch. With $\Lambda > 1$, the map $a \mapsto a + \phi\psi(a - a_{\text{sep}})$ is expanding away from the separatrix a_{sep}. The two fixed points on either side are attracting because the second-order terms in ϕ and ψ are negative (diminishing returns to addressability and to contribution). The result is bistability. □

Corollary 1 (The separatrix is a^*)**.** *Under the parameterisation of the first paper, the separatrix of the dynamic loop coincides with the static threshold a^*: the addressability at which per-drop contribution equals zero.*

This is the deepest result of the series. The threshold a^* that appeared as a static viability condition is not merely a static condition. It is **the boundary between the two attractors of the dynamic system**. A firm below a^* does not merely have unprofitable drops; it is being pulled, period after period, toward the death spiral. A firm above a^* is being pulled toward the high-density equilibrium. The boundary is the same; the interpretation is dynamical.

	Virtuous cycle	**Vicious cycle**
Starting state	Verified places, high-ρ clients	Unverified places, low-ρ clients
Consolidation	Routes fill, $G > 0$	Routes dilute, $G < 0$
Managerial response	Reinvest in ledger; sign repeatable clients	Cut price to buy volume
Effect on margin m	Rises (fewer failures, lower surcharge)	Falls (lower price, same failures)
Effect on $\rho = mR$	Rises → better selection	Falls → accept bad jobs to fill
Effect on consolidation	Better → G rises	Worse → G falls
Attractor	High-density equilibrium	Death spiral

The two cycles have the same equations and the same parameters. The only difference is the sign of the initial perturbation. **This is why capital could not save the previous generation.** Capital is a positive perturbation only if the loop gain is positive and the firm is above the separatrix. Below the separatrix, capital accelerates the spiral, because it lets the firm buy more volume that dilutes its routes further.

5 The Seven Components

Every paper in the series is an edge of the loop.

The loop is not a metaphor. It is a specific graph with specific edges. Each edge is a result in one of the prior papers. The synthesis is that the edges connect into a loop, and that the loop has two attractors.

#	Paper	What it establishes	Loop edge
1	Addressability Threshold	$a \to R(a)$: the rate primitive and its threshold a^*	The separatrix; the primitive
2	Address Regimes and Moat Durability	a^D compounds with volume; a^S does not	The feedback edge $G \to a$
3	The Drop Proof Rule	The six gates that test a single job	The per-job feasibility test
4	PURM (Pricing Engine)	The floor depends on N; the fixed point may not exist	The $N \leftrightarrow p$ interaction
5	Consolidation Constant	Load factor is the strategic variable	The $N \to G$ edge
6	Route Optimization	$\rho_j = m_j R(a_j)$ selects and orders jobs	The $\rho \to$ route composition edge
7	Bucket Algorithm	Assign jobs to future departures; kill below threshold	The operational realization of every edge
8	This paper	The loop, its two equilibria, its separatrix	The synthesis

6 The Bucket Threshold as a Drop-Rate Estimator

The loop's parameters are not directly observable. Addressability, margin, and the theoretical service rate are known only through the ledger and the scoring layer. But the **bucket system produces observable quantities that estimate them**.

6.1 The bucket as an observable

A bucket b has four observable properties at its cutoff time:

- **Load factor** $\lambda_{\text{obs},b} = N_{\text{obs},b}/K_v$: fraction of capacity filled.
- **Kill indicator** $\kappa_b \in \{0, 1\}$: whether the bucket was dispatched or killed.
- **Observed drops** $N_{\text{obs},b}$: parcels actually carried.
- **Observed duration** $T_{\text{obs},b}$: time to complete the route.

From these, two derived quantities connect bucket operations to theory.

Definition 4 (Observed service rate)**.** *The observed service rate of a bucket is*

$$R_{obs,b} = \frac{N_{obs,b}}{T_{obs,b} - \tau_{travel,b}}, \tag{16}$$

where $\tau_{travel,b}$ is the total travel time on the route.

Definition 5 (Utilisation ratio)**.** *The utilisation ratio of a bucket is*

$$u_b = \frac{R_{obs,b}}{R(a_b)}, \tag{17}$$

where $R(a_b)$ is the theoretical service rate computed from the scoring layer at the bucket's mean addressability.

6.2 The diagnostic

The utilisation ratio is the operational form of the loop diagnostic.

Proposition 2 (The utilisation diagnostic)**.** *For a bucket b in steady state:*

1. *If $u_b < 0.7$, the bucket is **consolidation-constrained**. The route runs at below the addressability-allowed rate, meaning demand is insufficient to fill it. The correct lever is client acquisition, corridor expansion, or a longer batching window.*
2. *If $u_b \in [0.7, 0.9]$, the bucket is in the **transition regime**. Either constraint could be binding; the kill rate is the tiebreaker.*
3. *If $u_b > 0.9$, the bucket is **addressability-constrained**. The route runs near the addressability-allowed rate, meaning demand is sufficient but service time is limiting. The correct lever is address verification, or raising drop value.*

This is the diagnostic that tells the firm which lever to pull. A firm observing $u < 0.7$ that invests in address verification is investing in the wrong lever — the address is not the bottleneck. A firm observing $u > 0.9$ that invests in client acquisition is investing in the wrong lever — the demand is not the bottleneck.

6.3 The kill rate as a density signal

The kill rate over a rolling window is the leading indicator of the corridor's health.

Definition 6 (Bucket kill rate)**.** *For a corridor over a rolling window W:*

$$\kappa_W = \frac{\#\{\textit{buckets killed in } W\}}{\#\{\textit{buckets scheduled in } W\}}. \tag{18}$$

Proposition 3 (Kill rate targets)**.** *In a healthy corridor:*

- *$\kappa_W \to 0$: buckets all fill, capacity is under-supplied relative to demand. Consider adding capacity or raising the kill threshold.*
- *$\kappa_W \approx 0.10$–0.25: healthy operating range. Most buckets fill above threshold; some are correctly killed rather than dispatched at a loss.*
- *$\kappa_W > 0.40$: corridor density is insufficient. Buckets are being killed faster than they are being filled. The corridor is either in the death spiral or has not yet reached viability.*

The kill rate is the operational read on the loop direction. A corridor with a rising kill rate is a corridor whose loop is running backwards. A corridor with a falling kill rate is a corridor whose loop is running forwards. The kill rate is observable in real time, and it precedes the P&L by weeks.

6.4 Estimating the drop rate from bucket observation

The drop rate the theory needs is $R(a)$, the theoretical maximum. The drop rate the firm actually achieves is R_{obs}. These differ by the utilisation ratio:

$$R_{\text{obs}} = u \cdot R(a). \tag{19}$$

This gives the operational estimator:

Proposition 4 (Drop rate estimator). *For a corridor with a stable bucket schedule, the achieved drop rate is estimated by*

$$\hat{R} = \bar{u} \cdot R(\bar{a}), \tag{20}$$

where $\bar{u}$ is the mean utilisation ratio over the rolling window and $\bar{a}$ is the mean addressability of dispatched places. The unachieved rate $R(\bar{a}) - \hat{R}$ is the consolidation gap: the portion of theoretical capacity that is lost to insufficient density rather than to poor addressing.

Corollary 2 (The consolidation gap is the strategic objective). *The strategic objective for each corridor is not to raise $\bar{a}$ or $\bar{u}$ independently, but to **close the consolidation gap** $R(\bar{a}) - \hat{R}$. This requires raising whichever of $\bar{a}$ or $\bar{u}$ is the binding constraint, as diagnosed by Proposition 2.*

This is why the bucket system is not just an operational convenience. It is the **measurement instrument** that turns the flywheel theory into a set of observable numbers on a dashboard.

7 Provider Assignment

The bucket system requires a rule for matching providers to filled buckets. The rule has three cases, and each is a specific solution to a specific contention problem.

7.1 The qualified and available sets

Definition 7 (Qualified providers). *$Q(b)$ is the set of providers qualified for bucket b: those with the required capacity class, corridor knowledge, and reliability score above threshold.*

Definition 8 (Available providers). *$A(b) \subseteq Q(b)$ is the subset of qualified providers currently undispached and within shift.*

7.2 Case 1: Multiple available providers

When $|A(b)| \geq 2$, contention exists and must be resolved.

Definition 9 (Provider performance score). *Each provider v has a performance score*

$$s_v = w_1 \cdot \textit{on-time rate} + w_2 \cdot \textit{first-attempt success} + w_3 \cdot \textit{data completeness} + w_4 \cdot (1 - \textit{complaint rate}), \tag{21}$$

with $\sum w_i = 1$ (default: $w = (0.35, 0.30, 0.20, 0.15)$).

The rule: weighted rotation.

$$v^* = \arg\max_{v \in A(b)} \left[\frac{s_v}{\text{dispatches}_{v,\text{recent}} + 1} \right]. \tag{22}$$

The numerator rewards performance. The denominator penalises recent dispatch, ensuring the rotation is fair. **The result is a rotation that is pegged to performance without being monopolised by the best provider.**

Why rotation and not auction? Because the bucket system relies on provider predictability. A provider who cannot predict which buckets they will be assigned will not invest in corridor-specific knowledge (the Thika Road traffic patterns, the specific gate guards, the recurring recipients). Rotation gives that predictability. The performance peg ensures it does not become a sinecure.

Why not pure dynamic assignment by cost? Because the firm does not observe provider cost directly. Cost is revealed by behaviour: a provider who accepts low-margin buckets has lower cost than one who does not. Rotation with a performance peg reveals this over time, at a much lower complexity than a full auction.

7.3 Case 2: Single available provider

When $|A(b)| = 1$, no contention exists. Assign immediately.

$$v^* = \text{the sole element of } A(b). \tag{23}$$

Why immediate and not delayed? Because there is nothing to optimise. The provider is available, the bucket is filled, the corridor needs the delivery. Delaying for a second provider would incur wait cost without a corresponding benefit.

The single-provider case is the diagnostic for a structural problem. If a corridor frequently has $|A(b)| = 1$, the corridor is **under-supplied on capacity**. The firm should either recruit more providers, accept lower reliability, or reduce the bucket frequency. The metric $|A(b)|$ is the observable signal.

7.4 Case 3: No available providers

When $|A(b)| = 0$ at cutoff:

- If the bucket's parcels can be re-routed to the next bucket in the corridor without violating their windows, **delay and re-bucket**.
- If they cannot, **kill the bucket** and route the parcels to the PREPARE state.

The kill response is correct here. A bucket with no provider is a bucket that cannot be dispatched profitably, regardless of how full it is.

7.5 The provider assignment rule in one line

Multiple providers → weighted rotation by performance. Single provider → immediate assignment. No provider → delay or kill. The rule resolves contention when it exists and does not create it when it does not.

8 The Operating Dashboard

The theory reduces to a small set of numbers that must appear on the operating dashboard daily.

Metric	Horizon	Target	What it diagnoses
Drops per vehicle-hour	Daily	Rising	North star; the whole loop
Utilisation ratio u	Per bucket	0.7–0.9	Which constraint binds
Bucket kill rate κ_W	Rolling 7d	0.10–0.25	Corridor health
Consolidation gap $R(\bar{a}) - \hat{R}$	Weekly	Falling	Strategic progress
Addressability $\bar{a}$	Weekly	Rising	Feedback edge working
Canonical places $n_j \geq 10$	Weekly	Rising	Schedulable base
Fragmentation rate	Weekly	Falling	Ledger quality
Premium share of revenue	Monthly	$\geq 30\%$	Tier discipline
$\lvert A(b)\rvert = 1$ frequency	Weekly	Falling	Capacity adequacy

The three numbers that matter most:

1. **Drops per vehicle-hour.** If this is rising, the loop is running in the virtuous direction. If it is flat or falling, the loop is stalling or reversing, and no other number can compensate.

2. **Utilisation ratio.** If u is below 0.7, the firm is consolidation-constrained and should acquire demand. If u is above 0.9, the firm is addressability-constrained and should verify places. The wrong response to the wrong constraint is the most common operational error.

3. **Bucket kill rate.** This is the earliest signal of corridor decay. A rising kill rate precedes a falling P&L by weeks and is the leading indicator of a developing death spiral.

Every other metric supports these three. None can substitute for them.

9 The Strategic Rules

The flywheel generates eight rules that are consequences of its structure, not independent strategic injunctions.

1. **Instrument every delivery.** The loop closes only if contribution converts to address data. A delivery without a ledger write is a courier job, not a moat deposit.

2. **Choose capacity for the corridor, not for the ambition.** K^* is determined by demand density. A vehicle larger than the corridor can fill is a subsidy.

3. **Never buy volume.** A price cut is a loop decision, not a margin decision. It accelerates the virtuous cycle only if the density elasticity exceeds the reciprocal of the density cost share. The fixed-point solver is the test.

4. **Diagnose before treating.** Read the utilisation ratio before investing in either address verification or client acquisition. The wrong lever is expensive.

5. **Client selection is the highest-leverage decision.** It determines $\bar{\rho}$, which determines route composition, which determines G, which determines whether the loop runs forward. Everything else is downstream.

6. **Verify the address before consolidating.** The three consolidation levers cannot be deployed without the address layer. Firms that tried to consolidate first failed for this reason.

7. **Pay per trip, not per parcel.** The rider's incentive must align with consolidation, which means the rider's fee must not rise with N.

8. **Watch the kill rate.** It is the leading indicator of the loop's direction, and it precedes the financials by weeks.

10 Falsifiability

The flywheel makes six predictions that would refute it.

1. **Against bistability:** evidence that firms in low-addressability markets with negative contribution eventually recover without external intervention. Would indicate no death spiral, only slow adjustment.

2. **Against the feedback edge:** evidence that firms with extensive ledgers show no addressability advantage over firms without. Would indicate the $G \to a$ edge is not real.

3. **Against the utilisation diagnostic:** evidence that firms responding to low u with address verification improve while firms responding with client acquisition do not (or vice versa). Would indicate the diagnostic is inverted.

4. **Against the kill rate targets:** evidence that corridors with kill rates above 40% are healthy, or that corridors below 10% are unhealthy. Would indicate the targets are miscalibrated.

5. **Against the separatrix coincidence:** evidence that the dynamical separatrix sits far from the static threshold a^*. Would indicate the static and dynamical analyses are decoupled.

6. **Against the loop gain:** evidence that a firm below the separatrix can be rescued by capital injection alone. Would indicate the loop gain is below 1 or has the wrong sign.

Each prediction is testable with data the firm generates during normal operation.

11 Relation to the Prior Literature

The flywheel is not a replacement for existing logistics theory. It is a superset.

- **The classical VRP** is the flywheel at $a = 1$ and $\Lambda = 0$: no addressability constraint, no feedback loop. It is a boundary case.
- **Load factor theory** (airlines) is the flywheel at $a = 1$ with $\Lambda > 0$: no addressability constraint, but the loop exists because capacity generates loyalty. The airline frequent flyer programme is the ledger.
- **Platform theory** is the flywheel viewed from the demand side: the loop closes through user acquisition rather than through delivery data.
- **Development economics** (infrastructure and growth) is the flywheel at the country level: investment feeds infrastructure feeds returns feeds investment. Same structure, different unit of analysis.

In each case, the flywheel reduces to a known theory at the boundary of its scope, and generalises it beyond that boundary. This is the correct relation of a general theory to the specific theories it contains.

12 Conclusion

Logistics is a flywheel. Address determines what is possible; the route determines what is attempted; consolidation determines what is realised; and realised contribution determines whether you earn the address data that makes more possible. The seven papers in this series are the seven edges of that loop.

The loop has two stable equilibria and a separatrix between them. The separatrix is the threshold that the first paper derived from static analysis. In the dynamical picture it is the boundary between the virtuous cycle and the death spiral. A firm is always on one side or the other. The question is not whether to be on the right side, but whether the firm knows which side it is on, and what would move it across.

The bucket system is the operational measurement instrument. Its utilisation ratio tells the firm which constraint is binding. Its kill rate tells the firm which direction the loop is running. Its provider assignment rule — weighted rotation for multiple providers, immediate assignment for one — resolves contention without creating it.

The dashboard reduces to three numbers: drops per vehicle-hour (the loop's speed), utilisation ratio (which constraint binds), and kill rate (which direction the loop is running). Every other metric supports these three, and none of them can substitute.

This is the synthesis. The seven papers are not seven ideas. They are one idea, with seven faces. The idea is that **logistics is a self-reinforcing loop, and the loop has a direction, and the strategic task is to be on the right side of it and to keep it turning.**

> **The theory in one sentence:** addressability sets the service rate, the service rate and margin set the selection, the selection sets the route, the route sets consolidation, consolidation sets contribution, and contribution determines whether you earn the address data that raises addressability — so logistics is a flywheel, not a funnel, and the entire strategic problem is to keep the flywheel turning in the right direction.

Acknowledgement of the series. This paper synthesises: *The Addressability Threshold and the Drop Rate* (the primitive and the separatrix), *Address Regimes and Moat Durability* (the feedback edge), *The Drop Proof Rule* (the per-job test), *The Peleka Universal Rate Model* (the pricing interaction), *The Consolidation Constant* (the load factor edge), *Route Optimization as Consolidation* (the selection rule), and *The Bucket Algorithm* (the operational realisation). Each is a component; this paper is the assembly.